

Clk	PC (Start)	Instruction	IR (Binary)	Opcode	Operation
0	0	$R1 \leftarrow \text{NSH}(R1)$	1000001	0	No-Shift (NSH)
1	1	$R1 \leftarrow R0 + R1$	1E+12	1	Add ($5 + 8 = 13$)
2	2	If $R0 > R1$ then branch to 3	1.111E+15	1111	Branch If Greater ($13 > 8$)
3	3	$R1 \leftarrow \text{ZERO}(R1)$	1.001E+09	0	Zero Register
4	4	If $ZF = 1$ then branch to 7	1.101E+15	1101	Branch If $ZF = 1$ (ZF is 1)
5	7	$R3 \leftarrow R7 \times R3$	1.01E+12	1	Multiply ($4 \times 6 = 24$)
6	8	$R1 \leftarrow R5 / R7$	1.011E+12	1	Divide ($36 / 4 = 9$)
7	9	If $R1 > R7$ then branch to 5	1.111E+15	1111	Branch If Greater ($9 > 4$)
8	5	$R2 \leftarrow \text{SHR}(R5)$	1.011E+11	0	SHR ($36 = 00100100_2$), $R5 \leftarrow 00010010_2 = 18$, $CF \leftarrow 0$, $R2 \leftarrow 18$
9	6	LAD ($R0 \leftarrow AC + 1$)	1.01E+15	1010	INC AC (AC holds $R5 = 36$), $AC \leftarrow 36 + 1 = 37$, $R5$ unchanged
10	10	$R0 \leftarrow R7 \times R3$	0101 1101	1	$R0 \leftarrow R7 \times R3$, $AC \leftarrow 4 \times 36 = 96$, $MQ \leftarrow 36$, $R0 \leftarrow 96$
11	11	$R2 \leftarrow R0 \text{ AND } R7$	1000 0011	1	$R2 \leftarrow R0 \text{ AND } R7$, $AC \leftarrow 96 \text{ AND } 4 = 0$, $MQ \leftarrow 4$, $R2 \leftarrow 0$
12	12	$R5 \leftarrow \text{ROR}(R5)$	1101 0100	0	$R5 \leftarrow \text{ROR}(R5)$, $AC \leftarrow 36$, $\text{ROR}(AC) = 18$, $R5 \leftarrow 18$
13	13	$R2 \leftarrow AC$	0000 0000	10	$R2 \leftarrow AC$, $AC = 18$, $R2 \leftarrow 18$
14	14	$AC \leftarrow R0$, $MQ \leftarrow R6$	0001 0111	11	$AC \leftarrow R0$, $MQ \leftarrow R6$, $AC = 96$, $MQ = 9$

Flags (SF, ZF, CF)	R0	R1	R2	R3	R4	R5	R6	R7	Next PC
0, 0, 0	5	8	12	6	16	36	9	4	1
0, 0, 0	5	13	12	6	16	36	9	4	2
0, 0, 0	5	13	12	6	16	36	9	4	3
0, 1 , 0	5	0	12	6	16	36	9	4	4
0, 1, 0	5	0	12	6	16	36	9	4	7
0, 0, 0	5	0	12	24	16	36	9	4	8
0, 0, 0	5	9	12	24	16	36	9	4	9
0, 0, 0	5	9	12	24	16	36	9	4	5
0, 0, 0	5	9	18	24	16	36	9	4	6
0, 0, 0	5	9	18	24	16	36	9	4	10
0, 0, 0	96	9	18	24	16	36	9	4	11
0, 1, 0	96	9	0	24	16	36	9	4	12
0, 0, 0	96	9	0	24	16	18	9	4	13
0, 0, 0	96	9	18	24	16	18	9	4	14
0,0, 0	96	9	18	24	16	18	9	4	15